Unified terpene pharmacology: a comprehensive evidence base

The same molecules—α-pinene, limonene, linalool, β-caryophyllene—are studied independently in cannabis pharmacology, aromatherapy, and forest medicine, yet researchers in each field rarely cite the others. This disciplinary fragmentation, rooted in the coincidence that both the Marihuana Tax Act and Gattefossé's *Aromathérapie* were published in 1937, has obscured a convergent pharmacological picture: terpenes act through shared molecular mechanisms regardless of botanical source, and the 2024 discovery of adenosine A2a receptor–mediated analgesia provides the mechanistic linchpin for reunification. Below is the complete evidence base—organized by the nine sections of the proposed review—with exact citations, quantitative data, and analytical notes on source quality.

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1. Terpene receptor pharmacology reveals a multi-target landscape far beyond CB1/CB2

β-Caryophyllene: the dietary cannabinoid

Gertsch et al. (2008, *PNAS* 105:9099–9104) identified (E)-β-caryophyllene (BCP) as a **full, selective CB2 agonist with Ki = 155 ± 4 nM** at the human CB2 receptor, using radioligand displacement of [^{3}H]CP55,940 in HEK293 cells. The (Z)-isomer showed lower affinity (Ki = 485 ± 36 nM). BCP displays no measurable CB1 binding, making it fully CB2-selective. Downstream signaling proceeds through Gi/Go coupling: dose-dependent inhibition of forskolin-stimulated adenylate cyclase ($R^2 > 0.97$), intracellular calcium transients in CB2-expressing HL60 cells ($R^2 = 0.99$, blocked by SR144528 at 1 μM), and weak activation of ERK1/2 and p38 MAPK in primary human monocytes. At 500 nM, BCP inhibits LPS-stimulated **IL-1β and TNF-α** expression in human peripheral whole blood, effects blocked by the CB2 antagonist AM630. In vivo, oral BCP at 5 mg/kg strongly reduces carrageenan-induced paw edema in wild-type mice but shows **no effect in CB2 knockout (Cnr2^{-}/$^{-}$) mice**, confirming CB2-dependent anti-inflammatory action.

Recent updates (2023–2025) extend BCP's mechanistic profile. BCP suppresses the TLR4/NF-κB/MAPK signaling cascade in diabetic cardiomyopathy models (ACS Pharmacol. Transl. Sci. 2023), abrogated by AM630. In LPS-stimulated HaCaT cells (Pharmaceuticals 2024;17:467), BCP at 10 μM reduced phospho-NF-κB p65, COX-2, and IL-1β, with effects reversed by AM630 at 100 nM. BCP+CBD combinations show synergistic cytokine suppression via dual CB2/PPARγ activation (Pharmaceuticals 2024). Additional targets include **TRPV1 and PPARγ**. No large-scale Phase II/III clinical trials of purified BCP have been reported as of early

2026, though Avicanna reported symptom improvements in musculoskeletal pain patients using BCP-containing formulations (April 2024).

d-Limonene: the A2a receptor agonist

Park et al. (2011, *Biochem. Biophys. Res. Commun.* 404:345–348) demonstrated that limonene exhibits **selective affinity for adenosine A2a receptors** in rat brain cerebral cortex radioligand binding assays. In CHO cells transfected with human A2a, limonene increased cytosolic cAMP, activated protein kinase A (PKA), phosphorylated CREB, and increased cytosolic calcium—the canonical A2a signaling cascade. Quantitative Ki values were not explicitly reported in this initial paper, but selective displacement was confirmed.

Siddiquee et al. (2021, *Purinergic Signalling*) confirmed limonene activates A2a (and possibly A2b) receptors, reducing airway inflammation in an ovalbumin-sensitized mouse asthma model (n=6/group); effects were absent in A2a knockout mice. Park et al. (2021, *J. Pharm. Pharmacol.*) demonstrated limonene anti-anxiety activity at 5–10 mg/kg IP in C57BL/6N mice, blocked by the A2a antagonist SCH58261, and mediated through regulation of dopaminergic and GABAergic neuronal function in the striatum.

Linalool: GABA-A modulation and the olfactory gate

Harada et al. (2018, *Front. Behav. Neurosci.* 12:241) provided the definitive anosmia experiment. Using n=240 male C57BL/6N mice (10/group), they showed linalool odor dose-dependently increased time in the light box and open arms of the elevated plus maze ($p<0.01$ to $p<0.0001$ vs. control). The **critical finding**: olfactory epithelium destruction with 3-methylindole (300 mg/kg IP, 2 weeks prior) **completely abolished** anxiolytic effects in anosmic mice. Anxiolysis was blocked by **flumazenil** (benzodiazepine-site antagonist) but NOT by WAY-100635 (5-HT1A antagonist, $p=0.7374$), confirming GABA-A–benzodiazepine receptor mediation through olfactory input rather than direct systemic action.

Milanos et al. (2017, *Front. Chem.* 5:46) provided electrophysiology data: whole-cell patch-clamp in HEK293 cells expressing α1β2γ2 GABA-A receptors showed linalool (2 mM) enhanced GABA-evoked currents to **160 ± 14% of GABA alone** ($p<0.05$). Linalool acts allosterically, independently of the γ2 subunit—meaning it does NOT bind the classical benzodiazepine site between α and γ subunits but rather a distinct allosteric site. Oxygenated metabolites retained activity; acetylated/carboxylated C8 derivatives lost it. Additional targets include TRPA1 inhibition and voltage-gated Ca^{2+} channel blockade in sensory neurons (Hashimoto et al. 2023, *Biochem. Biophys. Rep.* 34:101468), and glutamatergic modulation (Elisabetsky et al. 1995; Batista et al. 2008, *Neurosci. Lett.* 440:299–303).

α-Pinene: cholinergic enhancement and neurotrophin signaling

Acetylcholinesterase inhibition was quantified by Miyazawa & Yamafuji (2005, *J. Agric. Food Chem.* 53:1765–1768): **(+)-α-pinene IC_{50} = 0.40 mM; (−)-α-pinene IC_{50} = 0.44 mM** against bovine erythrocyte AChE. These are considerably weaker than pharmaceutical AChE inhibitors

(donepezil IC_{50} ~6–23 nM) but relevant at concentrations achievable via inhalation given α-pinene's **60% pulmonary bioavailability** (Falk & Hagberg 1990).

The BDNF/TrkB/CREB pathway was demonstrated by Hashemi et al. (2023, *Front. Mol. Neurosci.* 16:1202232): α-pinene (50 mg/kg/day IP, 21 days) in kainic acid–induced memory impairment in Wistar rats reversed decreases in hippocampal BDNF, TrkB, phospho-TrkB, CREB, and phospho-CREB by Western blot, with corresponding preservation of CA1 neurons and recovery of spatial/inhibitory avoidance memory. Kasuya et al. (2015, *Phytother. Res.* 29:43–47) showed inhaled α-pinene increased BDNF and tyrosine hydroxylase mRNA in mouse brain. In a Huntington's disease model (2024/2025), α-pinene at 1–10 mg/kg elevated BDNF, TrkA, TrkB, and CREB gene expression while decreasing p75NTR in cortex and hippocampus. Yang et al. (2016, *Mol. Pharmacol.* 90:530–539) identified α-pinene as a **partial modulator at the GABA-A benzodiazepine binding site**, enhancing non-REM sleep.

Myrcene: the paradox of CB1 engagement without direct agonism

Rao et al. (1990, *J. Pharm. Pharmacol.*) established that myrcene analgesia (10–40 mg/kg) involves **opioidergic** (naloxone-blocked) and **α2-adrenergic** (yohimbine-blocked) mechanisms, proposing α2-adrenoceptor–stimulated release of endogenous opioids. Myrcene also activates TRPV1 in a capsazepine-sensitive manner (Heblinski et al., *Channels*).

A striking recent finding: Alayoubi et al. (2025, *PAIN*, PMID: 40839768) showed myrcene (1–200 mg/kg IP) dose-dependently increased mechanical nociceptive thresholds in CCI neuropathic pain, with **greater potency in females**. CB1 antagonist blocked the effect—yet the TRUPATH assay confirmed myrcene **does NOT directly activate CB1** at 10 μM nor alter CB1 signaling by CP55,940, anandamide, or 2-AG. This "engagement without agonism" paradox suggests indirect mechanisms, possibly through endocannabinoid tone modulation or membrane effects. The "couch lock" sedation hypothesis at doses >0.5% in cannabis strains remains **largely anecdotal**; McKenna et al. (2022, *Int. J. Mol. Sci.*) specifically noted no effect on locomotion at analgesic doses.

1,8-Cineole, humulene, and minor terpenes

1,8-Cineole (eucalyptol) is primarily a **TRPM8 agonist** (EC_{50} = 120.4 μM at human TRPM8, 6-fold more potent than at mouse; Caceres et al. 2017, *Br. J. Pharmacol.*) and a **TRPA1 antagonist** (Takaishi et al. 2012, *Mol. Pain* 8:86). TRPM8 knockout abolishes its anti-inflammatory effects (Caceres 2017). α-Humulene shares sesquiterpene ancestry with BCP but **lacks CB2 affinity** (Ki >20 μM; Gertsch 2008). Its anti-inflammatory action proceeds via NF-κB and AP-1 inhibition, reducing TNF-α, IL-1β, PGE2, iNOS, and COX-2 at 50 mg/kg oral (Fernandes/Medeiros 2007, *Eur. J. Pharmacol.*). Among minor terpenes, geraniol is the most pharmacologically characterized: it acts as a strong TRPA1 modulator (EC_{50}/IC_{50} <1 μM for derivatives; Ortar et al. 2014), and was the **most effective** terpene in the Schwarz 2024 CIPN model. α-Bisabolol inhibits NF-κB via IκBα phosphorylation blockade and activates PPARγ (Kim et al. 2011). Terpinolene and ocimene remain pharmacologically poorly characterized with **no specific GPCR binding data published**.

TRP channel interactions: correcting a common mis-citation

De Petrocellis et al. (2011, *Br. J. Pharmacol.* 163:1479–94) is frequently cited as a terpene–TRP study but actually tested **phytocannabinoids** (CBD, THC, CBC, CBG, etc.) against TRP channels. Key cannabinoid–TRP data: CBC activated TRPA1 at EC_{50} = 90 nM; CBD antagonized TRPM8 at IC_{50} = 60 nM; CBD activated TRPV1 at EC_{50} = 1–3.5 μM. For actual terpene–TRP data, Caceres 2017, Takaishi 2012, and Heblinski et al. provide the primary evidence.

The Finlay null result reinterpreted through A2a

Finlay et al. (2020, *Front. Pharmacol.* 11:359) tested α-pinene, β-pinene, β-caryophyllene, linalool, and limonene individually and in mixtures at up to 10 μM in HEK293 cells expressing human CB1 or CB2. **No terpene displaced [^{3}H]CP55,940, altered THC/CBD binding, or modulated cAMP**. This was the key negative result arguing against a CB1/CB2-mediated entourage effect. However, LaVigne et al. (2021) found CB1-dependent ERK activation at higher concentrations using more sensitive assays, and Revel et al. (2023, *Biochem. Pharmacol.*, PMID: 37084981) detected all 16 tested terpenes activating CB1 at 10–50% of THC efficacy using the highly sensitive GIRK channel assay in Xenopus oocytes.

The resolution came from Schwarz et al. (2024, *PAIN*): terpene antinociception is mediated via **adenosine A2a receptors, not cannabinoid receptors**. The Finlay result was technically correct—terpenes show negligible direct CB1/CB2 orthosteric binding—but the question was wrong. The entourage effect does not require direct cannabinoid receptor activation; A2a provides the primary analgesic mechanism, while allosteric CB1 modulation may contribute secondarily.

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2. Inhalation pharmacokinetics demonstrate rapid CNS access and dose-dependent cognitive effects

Moss & Oliver 2012: the plasma–cognition correlation

Moss & Oliver (2012, *Ther. Adv. Psychopharmacol.* 2:103–113) exposed 20 healthy volunteers to rosemary essential oil aroma while performing cognitive tasks, then measured serum 1,8-cineole by GC-MS (terpinolene internal standard). Pearson correlations with serum concentrations revealed: serial threes accuracy **$r(18) = 0.469$, $p = 0.037$**; serial threes reaction time **$r(18) = -0.502$, $p = 0.024$**; serial sevens accuracy $r(18) = 0.433$, $p = 0.056$ (trend). RVIP accuracy showed no relationship, and contentedness showed a significant **negative** correlation, suggesting independent neurochemical pathways for cognition and mood. Effect sizes (r^2 = 0.19–0.25) were medium-to-large by Cohen's definitions. This was the **first demonstration that blood levels of an inhaled terpene correlate with cognitive performance** in humans.

Terpene PK parameters compiled from human studies

Jäger et al. (1996, *Chem. Senses* 21:477–480) established 1,8-cineole inhalation PK: **Tmax ~18 min, distribution t½α = 6.7 min, elimination t½β = 104.6 min** (~1.7 hours), with biphasic elimination and **2× longer half-lives in females** (attributed to subcutaneous fat as reservoir). Falk & Hagberg (1990, *Scand. J. Work Environ. Health* 16:372–378) documented α-pinene inhalation in 8 healthy males: **~60% pulmonary uptake**, distribution t½α ~5 min, elimination t½β = 26–38 min, and a third-phase t½γ = 695 min (~11.6 hours) reflecting high adipose affinity. D-Limonene shows **~65% pulmonary uptake** at 10–450 mg/m³ (Kohlert et al. 2000, *Planta Med.* 66:495–505). Most monoterpenes are rapidly metabolized to phase-II glucuronide conjugates via CYP3A4 (for 1,8-cineole) and eliminated renally, with only 1–3% eliminated unchanged via exhalation.

Blood-brain barrier penetration is rapid for all major terpenes

Monoterpenes (MW 136–154 Da, logP 2.7–4.8) easily satisfy BBB penetration criteria (<400 Da, logP >1.5). Direct mouse brain measurements (Satou et al. 2016, *Flavour Fragr. J.* 31:36–39) showed α-pinene reaching maximal brain levels within 30 min of inhalation (~90+ µg/g), limonene after 90 min (>50 µg/g), and linalool after 90 min. Critically, **α-pinene brain concentrations were 2-fold higher when inhaled as a mixed essential oil blend** versus single-compound inhalation, suggesting synergistic absorption. Pries et al. (2023, *Metabolites* 13:751) documented a BBB-opening effect of 1,8-cineole, and MacKenzie et al. (2023, *Sci. Rep.* 13:3605) provided the first demonstration of systemic distribution of orally administered 1,8-cineole to nasal polyp tissue in CRS patients.

Three nose-to-brain pathways

The nasal delivery framework (synthesized from Huang et al. 2023, *Frontiers*; Qiu 2025, *MedComm*; and multiple reviews) identifies three routes: (1) the **olfactory nerve pathway** (transcellular through sustentacular cells and perineural along olfactory nerve sheaths, through the cribriform plate to olfactory bulb); (2) the **trigeminal nerve pathway** (to brainstem via trigeminal branches, intact even in many anosmic patients); and (3) the **systemic respiratory pathway** (absorption through highly vascularized respiratory epithelium into bloodstream, then across BBB). For volatile monoterpenes at 136–154 Da, paracellular transport through tight junctions (<1 kDa optimal) and direct transcellular passage through lipophilic membranes are both available.

The Moss research program 2003–2025

Moss & Cook (2003, *Int. J. Neurosci.* 113:15–38): n=144 randomized to rosemary, lavender, or control. Rosemary enhanced Quality of Memory ($r^2 = 0.06$) and Secondary Memory ($r^2 = 0.07$); lavender impaired Working Memory. Moss et al. (2025, *Hum. Psychopharmacol.* 40:e70027): n=90 in double-blind positive-controlled trial using "Genius" essential oil blend (containing patchouli, neroli, grapefruit, cardamom, frankincense, spikenard, rosemary, lemongrass) with

fNIRS brain imaging. Both Genius and sage enhanced memory and executive function, with **Genius showing larger effects**. fNIRS revealed enhanced prefrontal cortex oxygenated hemoglobin during cognitive tasks, the first simultaneous demonstration of essential oil effects on both cognition and brain metabolism.

Schneider et al. (2019, *Hum. Psychopharmacol.* 34:e2683)

This paper reports across n=465 total participants using an AromaStick® inhaler delivering concentrated essential oils directly to one nostril. Physiological effects included: blood pressure changes, heart rate modulation (inhaler up to 3× more effective than room spray), **arterial O_2 saturation increased 2.5× above normal deep breathing**, salivary cortisol decreased, urinary serotonin increased, pain onset relief faster with duration and intensity reduced by up to **30%**, and visual scanning speed enhanced by up to **29%**. The author notes effects were "almost immediate, large, and clinically relevant." Important caveat: studies were funded by the AromaStick® manufacturer.

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3. Aromatherapy RCTs organized by terpene reveal a substantial but quality-limited evidence base

Linalool/lavender anxiolysis: 972 participants across 11 inhalation RCTs

Alhazmi et al. (2023, *Healthcare* 11:2978) systematically reviewed 11 RCTs (2018–2022) of lavender essential oil inhalation for anxiety, totaling **972 participants** (431 receiving lavender). Ten of 11 trials reported significantly decreased anxiety; populations included pre-surgical, hemodialysis, ECT, myofascial pain, IUI, postmenopausal depression, pediatric dental, and older adult settings. No pooled meta-analytic effect size was reported.

Kang et al. (2019, *Asian Nurs. Res.* 13:295–305) meta-analyzed 22 RCTs and found: anxiety reduction **Hedges' g = −0.65** (95% CI: −0.84 to −0.46, moderate-to-large effect), heart rate g = −0.53, systolic BP g = −0.22, **salivary cortisol g = −1.29** (large effect), and salivary chromogranin A **g = −2.29** (very large effect).

Donelli et al. (2019, *Phytomedicine* 65:153099) conducted the largest review: **102 RCTs + 25 non-randomized studies, >13,000 subjects**. Quantitative meta-analysis found oral lavender (80 mg/day, ≥6 weeks) reduced HAMA by MD −2.90 (95% CI: −4.86 to −0.95, p=0.004). Inhalation showed **Hedges' g = −0.73** (95% CI: −1.00 to −0.46, p<0.00001). However, the authors cautioned about "low average quality" and "high overall risk of bias," stating "it would be premature to consider lavender essential oil inhalation as a treatment for anxiety on the basis of our findings."

Silexan: the gold-standard anxiolytic terpene formulation

Kasper et al. (2014, *Int. J. Neuropsychopharmacol.* 17:859–869) conducted the pivotal trial: **N=539 adults with GAD**, randomized double-blind to Silexan 160 mg/day (n=121), Silexan 80 mg/day (n=135), paroxetine 20 mg/day (n=132), or placebo (n=135) for 10 weeks. HAMA reductions: Silexan 160 mg **−14.1 ± 9.3 points** ($p<0.01$ vs. placebo); Silexan 80 mg −12.8 ± 8.7 ($p<0.01$); paroxetine −11.3 ± 8.0 ($p=0.10$ vs. placebo in full analysis set). **Responder rates** (≥50% HAMA reduction): Silexan 160 mg 60.3%, Silexan 80 mg 51.9%, paroxetine 43.2%, placebo 37.8%. Remission (HAMA <10): Silexan 160 mg **46.3%** vs. paroxetine 34.1%. No withdrawal symptoms were observed for Silexan, unlike SSRIs. Adverse event incidence density was comparable to or lower than placebo.

Dold et al. (2023, *Eur. Arch. Psychiatry Clin. Neurosci.* 273:1615–1628) meta-analyzed all 5 completed Silexan DB-RCTs: **n=1,213**, Silexan 80 mg/day vs. placebo, 10 weeks. Responder rate ratio: **1.34** favoring Silexan; CGI much/very much improved rate ratio: **1.51**. Superiority observed across all diagnostic categories (subthreshold anxiety, mixed anxiety-depressive disorder, GAD). The Woelk & Schläfke (2010) trial showed Silexan 80 mg was comparable to lorazepam 0.5 mg for 6 weeks. Silexan is authorized in Germany as Lasea® and was included in Bandelow et al. (2023) clinical guidelines as recommended medication for anxiety.

1,8-Cineole/rosemary: cognitive domains enhanced

The Moss series establishes that rosemary aroma enhances: **Quality of Memory, Secondary (long-term) Memory** (Moss 2003); **working memory speed and accuracy** correlated with blood 1,8-cineole levels (Moss 2012); **prospective memory** (Moss 2017); and **prefrontal cortex metabolism** measured by fNIRS (Moss 2025). Effect sizes range from small (r^2 = 0.06–0.07 in Moss 2003) to medium-large (r^2 = 0.19–0.25 in Moss 2012). Juergens et al. (2003, *Respir. Med.* 97:250–256) demonstrated 1,8-cineole's anti-inflammatory potency in a respiratory context: oral 1,8-cineole 600 mg/day achieved **36% steroid dose reduction** in asthma patients vs. 7% placebo ($p=0.006$).

Limonene/citrus mood effects: pharmacology vs. hedonic valence

Watanabe et al. (2015, *Forsch. Komplementmed.* 22:43–49): n=41 healthy females in random crossover; bergamot essential oil (rich in limonene ~25–53%) significantly reduced salivary cortisol, increased parasympathetic HF-HRV ($p=0.003$), and improved negative mood. Komori et al. (1995, *Neuroimmunomodulation* 2:174–180): n=12 depressive patients (uncontrolled); citrus fragrance vaporization "markedly reduced" antidepressant doses and normalized NK activity and cortisol. This study is **frequently cited but methodologically weak**—no control group, no blinding, n=12. Importantly, Hoenen et al. (2016, *Front. Psychol.*) found d-limonene did NOT directly elevate mood vs. controls (n=78), but positive judgment of the odor was associated with mood improvement—suggesting hedonic valence may mediate citrus mood effects rather than direct pharmacology.

Estimated total participant count

Based on the Donelli 2019 aggregate (>13,000 across all lavender routes), O'Hara 2023 (4,681 across 50 inhalation RCTs for stress/anxiety), plus additional domains (pain, insomnia, cognition, nausea), the total across all published aromatherapy inhalation RCTs likely falls in the range of **15,000–25,000 participants**, with the Silexan program alone contributing 1,213 in high-quality DB-RCTs.

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4. The entourage effect: from endocannabinoid biochemistry to the A2a mechanism

Origin and evolution of the concept

Mechoulam & Ben-Shabat (1998, *Eur. J. Pharmacol.* 353:23–31) coined "entourage effect" in an **endocannabinoid** context: inactive 2-acyl-glycerol esters (2-linoleoyl-glycerol, 2-palmitoyl-glycerol) that co-occur with 2-AG in tissues potentiated 2-AG's CB receptor binding, adenylate cyclase inhibition, and in vivo tetrad effects, despite having no individual receptor activity. Russo's (2011, *Br. J. Pharmacol.* 163:1344–1364; cited **>2,000 times**) landmark "Taming THC" review extended this concept to phytocannabinoid-terpenoid synergy, proposing specific combinations for pain, depression, anxiety, epilepsy, cancer, and MRSA. Russo (2019, *Front. Plant Sci.* 9:1969) updated the argument, citing the Pamplona 2018 finding that CBD-rich cannabis extracts required only **22.5% of the pure CBD dose** (6.0 vs. 25.3 mg/kg/day) for epilepsy treatment.

Experimental evidence: the LaVigne–Schwarz–Spindle triad

LaVigne et al. (2021, *Sci. Rep.* 11:8232) tested α-humulene, geraniol, linalool, and β-pinene at 200 mg/kg IP in CD-1 mice (n=10–20/group). All produced cannabimimetic tetrad behaviors (analgesia, hypothermia, catalepsy, hypolocomotion). Rimonabant (CB1 antagonist, 10 mg/kg) **fully blocked** terpene-induced tail flick antinociception but did NOT reverse hypothermia. Istradefylline (A2a antagonist) blocked catalepsy. In vitro, all terpenes activated CB1-dependent ERK signaling. Terpene + WIN55,212-2 produced **enhanced, prolonged antinociception** beyond either alone.

Schwarz et al. (2024, *PAIN* 165:e145–e161) is the mechanistic breakthrough. Five terpenes (geraniol, linalool, β-pinene, α-humulene, β-caryophyllene) at 200 mg/kg IP produced antinociception **roughly equal to 10 mg/kg morphine** in CIPN, confirmed across four pain types including fibromyalgia and post-operative (Seekins et al. 2025, *Pharmacol. Rep.* 77:172–181). The A2a mechanism was demonstrated by: (1) istradefylline (3.2 mg/kg) significantly reducing antinociception; (2) **spinal cord–specific CRISPR knockdown of A2a** confirming the receptor; (3) in vitro cAMP modulation; (4) in silico molecular docking. Terpenes showed **no reward** (conditioned place preference), distinguishing them from opioids. Low-dose terpene (100 mg/kg) + morphine (3.2 mg/kg) produced **synergistic enhanced antinociception**.

Schwarz et al. (2025, PMID: 40122228) confirmed that natural cannabis terpene blends work via A2a, with **purified myrcene but NOT d-limonene** recapitulating antinociceptive effects—a crucial nuance for the pre-loading paradigm.

Spindle et al. (2024, *Drug Alcohol Depend.* 257:111267) provided the first controlled **human evidence**: 20 healthy adults, randomized double-blind within-subject crossover, vaporized d-limonene (1, 5, 15 mg) with THC (15, 30 mg). **30 mg THC + 15 mg d-limonene significantly reduced** ratings of "anxious/nervous" and "paranoid" vs. 30 mg THC alone in an orderly dose-response. Critically, limonene did NOT alter any other THC effect—not the "high," not cognitive impairment, not vital signs, and not THC pharmacokinetics. Limonene alone produced no pharmacodynamic effects. This selective anxiolysis represents the most direct human evidence for a specific terpene-cannabinoid interaction.

The dose gap and emerging resolution

Raz et al. (2024/2025, *Pharmaceuticals* 17:1543) acknowledged that animal study doses (200 mg/kg IP) are "substantially higher than those attainable through cannabis flower consumption." However, Raz et al. (2023, *Biochem. Pharmacol.* 214:115674) found that **16 cannabis terpenes each activated CB1 at 10–50% of THC efficacy** in a controlled heterologous expression system, with major amplification of THC activity occurring at terpene-to-THC ratios **similar to those found in the cannabis plant** and at concentrations as low as **0.001–0.01 μM**. This directly addresses the dose gap by showing synergistic CB1 effects at physiologically relevant, plant-ratio concentrations. The pre-loading paradigm—using external terpene sources (aromatherapy, forest exposure) to supplement cannabis terpenes—could further bridge this gap.

Pamplona et al. (2018, *Front. Neurol.* 9:759)

Meta-analysis of 11 observational studies, 670 patients with treatment-resistant epilepsy. CBD-rich cannabis extracts showed clinical improvement in 71% vs. 46% for purified CBD. The **4× potency difference** (6.0 vs. 25.3 mg/kg/day average dose) was accompanied by dramatically fewer adverse events: mild AEs in 33% (CBD-rich) vs. 76% (purified, $p<0.001$); severe AEs in 7% vs. 26% ($p<0.0001$). Limitation: observational, non-blinded design.

Ferber et al. (2020, *Curr. Neuropharmacol.* 18:87–96)

Review proposing terpene-cannabinoid combinations for mood and anxiety disorders. Highlighted that d-limonene (1% inhaled) showed anxiolytic effects NOT blocked by flumazenil, indicating a mechanism distinct from benzodiazepines. Called for innovative terpene-cannabinoid formulations for psychiatric conditions.

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5. Forest bathing immunology: phytoncides as the mechanism behind shinrin-yoku

Qing Li's NK cell research program

Li et al. (2007, *Int. J. Immunopathol. Pharmacol.* 20(2 Suppl 2):3–8): 12 healthy males, 3-day/2-night forest trip. **NK activity increased ~50%**; 11/12 subjects showed elevated NK activity. Significant increases in NK cell numbers and intracellular perforin, granulysin, granzymes A/B. Li et al. (2008a, *Int. J. Immunopathol. Pharmacol.* 21:117–128): 12 males in Chamaecyparis obtusa forest vs. Nagoya city control. Forest trip increased NK activity, NK cells, anti-cancer proteins, and **decreased urinary adrenaline**. City trip produced no changes. **Phytoncides α-pinene and β-pinene were detected in forest air** but were absent in city air. Li et al. (2008b, *J. Biol. Regul. Homeost. Agents* 22:45–55): 13 female nurses, same protocol, confirming results across sexes.

Duration of immune enhancement: Individual studies showed effects lasting **>7 days**. Li's 2010 comprehensive review (*Environ. Health Prev. Med.* 15:9–17) stated: **"The increased NK activity lasted for more than 30 days after the trip, suggesting that a forest bathing trip once a month would enable individuals to maintain a higher level of NK activity."**

The in vitro phytoncide study (Li et al. 2006, *Immunopharmacol. Immunotoxicol.* 28:319–333) tested **α-pinene, 1,8-cineole, and d-limonene** directly on NK-92MI cells (incubated 48–144 hours). These phytoncides dose-dependently increased cytolytic activity and intracellular perforin, granzyme A, and granulysin. Kim et al. (2021, *Int. J. Mol. Sci.* 22:656) extended this: α-pinene at 100 μM increased perforin, granzyme B, CD56, and CD107a in NK-92mi cells via ERK/AKT signaling, and inhibited CT-26 tumor growth in vivo.

The controlled hotel room experiment (Li et al. 2009, *Int. J. Immunopathol. Pharmacol.* 22:951–959) is the strongest evidence isolating the phytoncide variable: 12 males stayed in an urban hotel with hinoki cypress essential oil vaporized via humidifier overnight. **NK activity, anti-cancer proteins, and intracellular markers all increased significantly, while urinary catecholamines decreased**—replicating forest bathing effects in a controlled indoor setting. The phytoncide meta-analysis by Piva et al. (2024) yielded an NK cell activation **effect size of 2.50** (95% CI: 1.94–3.05, $p<0.05$, $I^2=50.47\%$)—a very large effect.

Li's 2022–2025 expansion

Li et al. (2022, *Environ. Health Prev. Med.* 27:44) demonstrated forest bathing significantly increased **blood serotonin** and reduced depression scores—the first serotonin measurement in this context. Li et al. (2025, *Diseases* 13:100): 31 women with depression showed decreased SDS scores, increased serotonin, **oxytocin, and IGF-1**, with effects persisting one week. Li et al. (2025, *J. Occup. Health* 67:uiaf041): 30 males with COPD risk showed reduced inflammatory markers and improved SpO_2 after forest bathing in a randomized crossover design—the first COPD application.

Shared molecules across three settings

Forest air contains total terpenes at **3.5–56.0 μg/m³**, with α-pinene reaching **up to 13.05 μg/m³** in pine forests and limonene at **0.81–8.10 μg/m³**. These same molecules dominate cannabis terpene profiles (α-pinene 0.3–1.5%, limonene 0.4–1.5% of dry weight) and essential oil diffusion. The concentration difference is enormous—forest levels are 1,000–100,000× lower than essential oil diffuser or cannabis vapor settings—yet biological effects remain demonstrable, potentially through olfactory signaling rather than systemic receptor engagement. The NHANES 2013–2014 population study (n=2,213) found baseline serum limonene of **~1,200 ng/L** and α-pinene of **~76 ng/L** in the US population, with citrus consumption increasing serum limonene by 82.1%.

The most methodologically rigorous forest terpene study to date—Levy et al. (2025, *Environ. Res.* 276:121482), a double-blind randomized crossover filtering forest air terpenes—found **no consistent differences** in psychophysiological or immunological outcomes between filtered and unfiltered 60-minute forest air sessions, suggesting longer exposure or multi-sensory integration may be necessary to replicate effects documented in multi-day studies.

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6. The A2a-CB1 heteromer system provides the molecular substrate for terpene-cannabinoid crosstalk

Architecture and location of the heteromer

Moreno et al. (2018, *Neuropsychopharmacology* 43:964–977) used in situ proximity ligation assay with conditional CB1 knockout mice to show A2a-CB1 heteromers are **essentially absent from corticostriatal projections and striatonigral (direct pathway) neurons** but **largely present in striatopallidal (indirect pathway) neurons**. Co-stimulation of both receptors produces **cross-antagonism**—strongly reduced ERK and Akt phosphorylation—and the heteromer selectively couples to **Gq proteins** rather than the individual protomers' canonical Gs (A2a) and Gi (CB1).

Kőfalvi et al. (2020, *BMC Biol.* 18:9) established the quaternary structure as an **$(A2a)_2$-$(CB1)_2$ heterotetramer**, dependent on intermolecular interactions involving the long A2a C-terminus, which determines constitutive adenylyl cyclase activation via Gs. G-protein coupling specificity within the heteromer: NECA (A2a agonist) EC_{50} = **0.35 ± 0.08 μM (Gs only)**; CP55940 (CB1 agonist) EC_{50} = **1.05 ± 0.39 μM (Gi only)**. The critical finding: CB1 agonist–mediated inhibition of striatal glutamate release primarily occurs through counteraction of A2a's constitutive adenylyl cyclase activation—meaning the well-established cannabinoid inhibition of glutamate release depends on the A2a-CB1 heteromer architecture.

Hippocampal heteromers and CBD's mechanism

Aso et al. (2019, *Mol. Neurobiol.* 56:5382–5391) revealed A2a-CB1 heteromers at the **presynaptic level in hippocampal CA1 neurons** and demonstrated that CBD blunts THC-induced cognitive impairment in an A2a-dependent manner. CBD was NOT capable of modifying motor functions or anxiety—only cognition—demonstrating **brain region–dependent functional specificity** of the heteromer.

Sánchez-Fernández et al. (2024, *Acta Neuropsychiatr.* 36:320–324) provided the molecular mechanism: CBD acts as a **negative allosteric modulator of A2a**, decreasing agonist (CGS21680) potency 3.5–5 fold without acting as an agonist itself. Raïch et al. (2023, *Int. J. Mol. Sci.* 24:17500) confirmed CBD at 200 nM decreased A2a-downstream ERK phosphorylation by ~40%. This provides a mechanistic basis for the Aso finding: CBD modulates THC's cognitive effects by negatively modulating A2a within the heteromer complex.

A2a antagonism prevents cannabinoid-induced memory deficits

Mouro et al. (2017, *Neuropharmacology* 117:316–327; 2019, *Neuropharmacology* 155:10–21) showed that chronic istradefylline (A2a antagonist, 3 mg/kg/28 days) **reverted recognition memory deficits** induced by chronic WIN55,212-2 (1 mg/kg/28 days). At the synaptic level, WIN at 300 nM impaired hippocampal CA1 LTP; SCH 58261 at 100 nM **partially rescued** this impairment. Chronic administration of neither agent altered A2a or CB1 receptor density in hippocampus or prefrontal cortex.

The pre-loading mechanism at the heteromer

Based on the heteromer pharmacology, **activating A2a first** (e.g., via limonene or other A2a agonist terpenes) would: (1) increase adenylyl cyclase activity through Gs; (2) establish an allosteric state reducing subsequent CB1 Gi-mediated signaling efficiency; (3) reduce THC's ability to inhibit glutamate release at corticostriatal terminals; and (4) at postsynaptic striatopallidal neurons, produce cross-antagonism (reduced ERK/Akt). The net effect would be selective attenuation of certain CB1-mediated effects. However, this is brain region–specific: hippocampal A2a-CB1 heteromers mediate cognitive effects while striatal heteromers mediate motor effects (Moreno 2018). Ferré et al. (2023, *Trends Pharmacol. Sci.* 44:495–506) formalized three types of allosteric interactions within the heteromer: Type I (ligand-dependent), Type II (ligand-independent/constitutive), and Type III (through plasma membrane effectors).

Chiodi et al. (2016, *J. Neurochem.* 136:907–917) used transgenic rats overexpressing A2a to show that CB1 agonist effects were significantly reduced, while Carlin et al. (2018, *Neuropharmacology* 139:268–278) demonstrated A2a agonism causes hypothermia via peripheral receptors—connecting to the tetrad. No cryo-EM structure of the A2a-CB1 heteromer exists as of early 2026; the Kőfalvi computational model remains the best structural description. The A2a-CB2 heteromer has also been recently characterized (Llinas Del Torrent 2024/2025, *Br. J. Pharmacol.*; Rivas-Santisteban 2025), with CBD acting as a biased allosteric modulator.

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7. Network pharmacology provides the intellectual framework for cross-botanical terpene design

Foundational concepts

Hopkins (2008, *Nat. Chem. Biol.* 4:682–690; >3,310 citations) coined "network pharmacology," arguing that biological robustness makes single-target drugs inherently limited and that modulating multiple network nodes offers superior efficacy. Wagner & Ulrich-Merzenich (2009, *Phytomedicine* 16:97–110; >1,495 citations) identified four synergistic mechanisms in phytopharmaceuticals: multi-target effects, pharmacokinetic enhancement, bioavailability improvement, and resistance mechanism circumvention. Gertsch (2011, *Planta Med.* 77:1086–1098) explicitly applied these principles to terpenoids, proposing botanical drugs as "intelligent mixtures" functioning as inherent network pharmacology agents. Anighoro et al. (2014, *J. Med. Chem.* 57:7874–7887) formalized polypharmacology as the design of drugs modulating multiple targets, with Designed Multiple Ligands (DMLs) as an emerging paradigm—terpenes, with their moderate affinity across multiple targets, represent natural DMLs.

Cross-plant enhancement precedents

The piperine-curcumin paradigm is the premier example: Shoba et al. (1998, *Planta Med.* 64:353–356) showed piperine from black pepper increased curcumin bioavailability by **154% in rats and 2,000% in humans** by inhibiting hepatic/intestinal glucuronidation. Traditional Chinese Medicine provides millennia of precedent through the Jun-Chen-Zuo-Shi combinatorial framework, with 1,444 multi-herb formulas in the Chinese Pharmacopoeia (Zhou et al. 2016, *Front. Pharmacol.* 7:201). Systems pharmacology analysis of TCM formulas (Li et al. 2018, *Sci. Rep.* 8:380) confirmed that different compounds from different source organisms share disease-relevant targets enriched in the same pathways.

Mathematical optimization frameworks

Isobolographic analysis (Tallarida 2006, *JPET* 319:1–7) uses dose equivalence to identify synergy (below the isobole) vs. antagonism. For terpenes acting on different receptor targets, Bliss Independence (1939) may be more appropriate than Loewe Additivity (drugs sharing the same target). The Chou-Talalay Combination Index (Chou 2006, *Pharmacol. Rev.* 58:621–681) provides a unified framework using the median-effect equation. Network-based drug combination prediction (Cheng et al. 2019, *Nat. Commun.* 10:1197) uses human interactome proximity to predict effective combinations—directly applicable to terpene blend optimization.

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8. Funding landscape faces existential threat despite emerging translational momentum

Federal mechanisms

The NCCIH Notice of Special Interest NOT-AT-22-027 is now **expired** but spawned ~$3 million+ in grants across the initial RFA round (9 awards). It listed 10 specific terpenes: myrcene, β-caryophyllene, limonene, α-terpineol, linalool, α-phellandrene, α-pinene, β-pinene, γ-terpinene, and α-humulene. Key funded investigators include Sara Jane Ward (Temple), Jenny Wiley (RTI), Cassandra Quave (Emory)/Isaac Chiu (Harvard), John Streicher (Arizona), Ziva Cooper (UCLA), and David Sarlah (UIUC).

Streicher's R01AT011517 (NIH/NCCIH) remains **active**, funding the LaVigne→Schwarz→Seekins progression that discovered the A2a mechanism. Streicher consults for Napreva (terpene product company) and holds equity in Botanical Results and Teleport Pharmaceuticals.

The **R3CR** (Resource Center for Cannabis and Cannabinoid Research) launched April 2025 under grant U24AT013161 (NCCIH, with NCI/NIA/NIDA co-funding), hosted at University of Mississippi with WSU and USP partnerships. Three cores: Regulatory Guidance, Research Support (including seed funding), and Research Standards. First seed funding proposals were due December 8, 2025.

Critical threat: The FY2026 White House budget proposes **complete elimination of NCCIH**—the primary NIH funder of terpene research. By mid-2025, ~2,300 NIH grants ($3.8B+) had been terminated or frozen. While Congress has pushed back, the outcome remains uncertain as of February 2026. The NIDA cancellation of the 57-year University of Mississippi cannabis production contract further destabilizes the infrastructure.

Johns Hopkins BPRU

The Cannabis Science Laboratory (Vandrey, Spindle) operates under NIDA P50DA056408 and R01DA043475. Beyond the Spindle 2024 limonene study, Kumar et al. (2025, *Med. Cannabis Cannabinoids*) tested α-pinene + THC (n=19), finding **no attenuation of THC-induced memory impairment**—an important negative result. Jenkins et al. (2023, *Cannabis Cannabinoid Res.*) tested 5 terpenes preclinically, finding bisabolol and d-limonene showed anxiolytic effects. Planned studies include oral d-limonene replication and myrcene sedation investigation. Patent application PCT/US2022/014296 has been filed for d-limonene to reduce THC-induced anxiety.

California DCC grants

The California DCC has invested nearly **$80 million** since 2020 across three rounds. The 2025 round awarded ~$30 million for 22 projects. Terpene-relevant awards include UCLA's "Flower

Flavor-Compound Reference Dataset" ($1.23M) and San Diego State's cannabis worker terpene exposure assessment ($1.16M).

Regulatory advantage

Most individual terpenes are **GRAS** (FDA 21 CFR 172.510 / FEMA evaluations). β-Caryophyllene received explicit GRAS status. Hemp-derived terpenes are unregulated under the 2018 Farm Bill. The FDA IND pathway has been successfully navigated (IND#140339 for Hopkins terpene studies). On December 18, 2025, **Executive Order 14370** directed the Attorney General to expedite cannabis reclassification to Schedule III, though the rulemaking timeline remains uncertain.

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9. The 1937 coincidence created three parallel disciplines studying identical molecules

The year of divergence

In a remarkable historical coincidence, **1937 saw both the Marihuana Tax Act and the publication of Gattefossé's *Aromathérapie***. The Tax Act, championed by FBN Director Harry Anslinger, imposed prohibitive taxes and bureaucratic barriers that "effectively ended most cannabis research in the United States." Cannabis was removed from the US Pharmacopoeia in 1942. Meanwhile, Gattefossé's book formalized the scientific study of the same terpenes (linalool, pinene, limonene) under the entirely separate banner of aromatherapy. Shinrin-yoku was coined in 1982 by Tomohide Akiyama of the Japanese Ministry of Agriculture, Forestry and Fisheries, and Qing Li began formal research in 2004.

The **regulatory architecture created the silos**: cannabis terpenes required Schedule I authorization to study (DEA registration, sole-source material from University of Mississippi until 2021, FDA IND, state approvals); the identical molecules from lavender, pine, or citrus required no special authorization. The result was three parallel literatures: cannabis pharmacology (journals: *Cannabis Cannabinoid Res.*, *J. Cannabis Res.*), aromatherapy (*Int. J. Aromather.*, *Complementary Therapies*), and forest medicine (*Environ. Health Prev. Med.*, *Int. J. Environ. Res. Public Health*). No published bibliometric study quantifying cross-citation rates between these fields was found, but the structural isolation is evident: Qing Li frames α-pinene and limonene as "phytoncides," Russo as "cannabis terpenes," and Moss as "essential oil constituents"—nomenclature reflecting disciplinary boundaries, not chemistry.

Market forces reinforced the divide

The essential oil industry grew to **$26.91 billion globally in 2024** (Polaris Market Research), led by companies like doTERRA (founded 2008, 5 million customers) and Young Living

(founded 1993). These companies freely commercialized limonene, linalool, and pinene—the same molecules that cannabis researchers needed Schedule I authorization to study. The paradox is stark: identical molecules are GRAS food additives and unrestricted essential oil ingredients, yet Schedule I when present in cannabis with >0.3% THC.

Convergence signals

The Leafly cannabis guide (~2019–2020) abandoned the indica/sativa/hybrid taxonomy in favor of **terpene-based classification**, using 8 color-coded terpenes. The Emerald Cup (2021) began sorting flower entries by **six primary terpene groups** (myrcene, β-caryophyllene, limonene, terpinolene, α-pinene, β-ocimene), with SC Labs' Alec Dixon declaring "the real quantitative quality marker in cannabis is terpene concentration." The Jikomes et al. (2022, *PLOS ONE*) analysis of 89,923 commercial samples confirmed that terpene profiles—not indica/sativa labels—are the key chemical differentiators.

The psychedelic research renaissance provides a parallel: ~30 years of dormancy after prohibition, followed by explosive growth (54% of top-100-cited psychedelic papers published 2010–2020). However, terpene fragmentation is **disciplinary** (same molecules, parallel fields) rather than primarily temporal (same field, interrupted), making reunification conceptually harder but potentially more impactful.

Conclusion: the evidence supports a unified framework with critical caveats

The evidence base for a unified terpene pharmacology is substantial and growing. The A2a receptor mechanism (Schwarz 2024) resolves the Finlay 2020 null result and provides a non-cannabinoid pathway through which terpenes exert analgesia equal to morphine in preclinical models. The A2a-CB1 heteromer system (Moreno 2018, Kőfalvi 2020, Aso 2019) provides the molecular substrate for terpene-cannabinoid crosstalk, with brain region–specific functional consequences. The Spindle 2024 human trial demonstrates selective anxiolysis in the first controlled evidence of a specific terpene-cannabinoid interaction.

Three critical unresolved issues deserve emphasis. First, the **dose gap** between animal studies (200 mg/kg IP) and human cannabis consumption remains the field's central challenge, only partially addressed by Raz et al.'s (2023) finding of CB1 synergy at plant-relevant concentrations in vitro and the pre-loading concept of supplementing cannabis terpenes with external sources. Second, the **inhalation route did not produce terpene antinociception** in the Schwarz mouse model—a significant limitation for translational relevance to vaporized cannabis. Third, the **federal funding infrastructure faces existential threat** from the proposed elimination of NCCIH, the primary NIH sponsor of this research domain.

The pre-loading paradigm—using aromatherapy or forest medicine terpene exposure to prime A2a receptors before cannabis-derived CB1 activation—represents a testable hypothesis that would operationalize the cross-botanical framework. With A2a-CB1 heteromers concentrated in striatopallidal neurons and hippocampal CA1 synapses, the molecular machinery exists. Whether the 1,000–100,000× concentration gap between forest air terpenes and pharmacologically active doses can be bridged through olfactory signaling, prolonged low-dose exposure, or supplementation with essential oil inhalation remains the key empirical question for this field's next decade.